

Generalized Bonferroni and Minimax Error Spending

Honors in Mathematical and Data Sciences Research

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Abstract

This research examines how to allocate Type I and Type II familywise errors across multiple endpoints when all tests must be satisfied by the same sample size. The central question is whether equal Bonferroni splitting wastes observations when endpoint effect sizes differ, and if so, how much of that loss can be recovered by a more efficient allocation rule.

For d tests with effect sizes $\delta_1, \dots, \delta_j$, the required sample size for endpoint j depends on the assigned error probabilities α_j and β_j and the overall study size is determined by the largest endpoint requirement. Three allocation rules are compared: standard Bonferroni, proportionate spending, and minimax optimal spending.

The numerical study varies the effect size ratio $\rho = \delta_1 / \delta_2$ and evaluates how allocation patterns and sample size savings change as endpoint heterogeneity increases. The main finding is that equal Bonferroni splitting is only optimal when effect sizes are equal. Once endpoints differ in difficulty, the minimax rule shifts more α and β toward the harder test, balancing endpoint requirements, and reducing the overall required sample size. In a worked example, the required sample size falls from 263 under Bonferroni to 164 under minimax, with minimax having the smallest sample size while preserving the same total error budgets.

1. Introduction

Modern experiments, especially clinical trials, often evaluate several endpoints on the same group of participants (O'Brien & Fleming, 1979; Dmitrienko, Tamhane, & Bretz, 2010). In this setting, the study cannot be sized separately for each endpoint so instead, the final sample size must be large enough for all tests simultaneously (Dmitrienko, Tamhane, & Bretz, 2010). As a

result, the way error is distributed across tests directly affects the required sample size and plays an important role in determining how efficiently the study can be designed.

The common standard Bonferroni approach divides the total Type I error and Type II error budgets equally across endpoints (Bonferonni, 1936). This rule is simple and guarantees familywise control, but it ignores differences in endpoint difficulty. When one endpoint has a smaller effect size than another, equal splitting can force the hardest endpoint to dominate the required sample size and can therefore waste observations (O'Brien & Fleming, 1979).

This project studies whether a more precise allocation of error probabilities can reduce that waste. The work builds on Dr. Michael Baron's framework for generalized Bonferroni and minimax error spending, supported by numerical computations carried out using custom R code functions `BonfErrorSpending()`, `PropErrorSpending()`, and `OptErrorSpending()` (Baron, 2025; Baron & MacMillan, 2019). The main objective is to compare equal splitting, proportionate spending, and minimax optimal spending, and to determine when optimal reallocation reduces the required overall study size.

Four core ideas guide this report. First, equal splitting is optimal when all effect sizes are equal. Second, when tests differ in difficulty, the harder test should receive a larger share of both α and β under the minimax rule. Third, this redistribution lowers the maximum endpoint specific sample size and therefore lowers the overall required N . Fourth, the gains from optimal allocation increase as the heterogeneity of endpoint effect sizes increases.

2. Problem Setup and Notation

Suppose d endpoints are tested using the same sample. Let ρ_j reflect the effect size for endpoint j and let α_j and β_j reflect the Type I and Type II error probabilities assigned to that endpoint.

The required sample size for test j is written as:

$$n_j = \left(\frac{z_{\alpha_j} + z_{\beta_j}}{\delta_j} \right)^2$$

The overall study size must satisfy every endpoint requirement, so the design is determined by the largest of these values:

$$N = \max_j n_j$$

Therefore, the reallocation is not just to control familywise error, but to distribute the total error budgets in the most efficient way. If one endpoint is significantly harder than the others, then giving every test the same amount of α and β may be unnecessary.

To measure heterogeneity between the two endpoints in the numerical study, we use the ratio $\rho = \delta_1 / \delta_2$. When $\rho = 1$, the endpoints are equally difficult. Values of ρ far from 1 indicate increasing imbalance, meaning one test is easier and the other is harder. This research examined how the error shares and the resulting sample size savings behave as ρ moves away from 1.

3. Error Spending Rules

This research compares three ways to allocate the total Type I and Type II error budgets across endpoints.

3.1 Standard Bonferroni splitting

Under standard Bonferroni spending, the error budgets are divided equally across all d endpoints.

This gives $\alpha_j = \frac{\alpha}{d}$ and $\beta_j = \frac{\beta}{d}$ for every j . The method is transparent and easy to implement and treats easy and difficult tests as if they were equal.

3.2 Proportionate spending

The proportionate rule allocates error according to endpoint difficulty through constants C_α and C_β in the notation:

$$\alpha_j = \Phi(-C_\alpha \delta_j), \quad \beta_j = \Phi(-C_\beta \delta_j)$$

This rule is more adaptive than equal splitting because it typically assigns more error to harder endpoints. In earlier stages of the project, this method served as a practical reference point because it shows the main behavior of the minimax solution while remaining easier to compute.

3.3 Minimax optimal spending

The minimax rule determines the error allocations by minimizing the largest endpoint specific sample size, subject to the total Type I and Type II error constraints. Formally, the problem is:

$$\min_{\{\alpha_j, \beta_j\}} \max_j n_j$$

Subject to:

$$\sum_{j=1}^d \alpha_j = \alpha, \quad \sum_{j=1}^d \beta_j = \beta$$

This method is particularly best in the shared sample setting because the overall study size is determined by the most difficult endpoint. Rather than allowing a single difficult test to control the design, the minimax rule redistributes error probabilities so that endpoint requirements are balanced as evenly as possible, therefore reducing the total sample size needed to satisfy all tests simultaneously.

4. Minimax Allocation System

The R function `OptErrorSpending()` solves numerically for a constant C and the overall required sample size N . The full system is:

$$\alpha = \sum_{j=1}^d \Phi\left(-\frac{\delta_j^2 N + C}{2\delta_j \sqrt{N}}\right) \quad \beta = \sum_{j=1}^d \Phi\left(-\frac{\delta_j^2 N - C}{2\delta_j \sqrt{N}}\right)$$

Once C and N are determined, the endpoint specific minimax allocations are found as:

$$\alpha_j = \Phi\left(-\frac{\delta_j^2 N + C}{2\delta_j \sqrt{N}}\right), \quad \beta_j = \Phi\left(-\frac{\delta_j^2 N - C}{2\delta_j \sqrt{N}}\right)$$

These systems and formulas make the balancing method beneath the minimax rule clear. If all effect sizes are equal, the minimax solution falls back to equal splitting. If one test is harder, meaning it has a smaller effect size, then the minimax rule shifts more α and β towards that endpoint so that the largest required sample size is reduced and therefore lowers the overall study size required to satisfy all tests.

Once C and N are found simultaneously as part of the minimax solution, the resulting N represents the required overall study size under minimax allocation. The endpoint specific allocations α_j and β_j are then generated directly from the solution that determines the smallest feasible overall N .

5. Asymptotic Result

A derived asymptotic result from Dr. Micheal Baron shows that Bonferroni splitting carries an unavoidable additive sample size cost when multiple endpoints are tested simultaneously (Baron, 2025). Specifically, as $\alpha, \beta \rightarrow 0$,

$$N_{\text{BONF}} - N_{\text{no-adjust}} \approx \frac{2 \log(k)}{\delta^2}$$

So although the relative cost of Bonferroni splitting becomes small in extremely low error rates, the absolute cost remains positive. This motivates studying whether optimal error allocation can recover some of that loss when endpoints differ in difficulty. In other words, even when the penalty of equal splitting appears small on a relative scale, it can still lead to additional participants that may be unneeded through more efficient error allocation.

6. Numerical Study Design

The numerical study fixes the total Type I and Type II error budgets α and β , varies the effect size ratio $\rho = \delta_1 / \delta_2$, and compares the overall required sample size under the three allocation rules.

For each value of ρ , the overall requirement is computed as:

$$N = \max_j n_j$$

The main results are the allocation of Type I error, the allocation of Type II error, the proportional saving relative to standard Bonferroni, the absolute sample size saving, and the direct comparison of required overall sample size across methods. This design allows us to study both the mechanism of the allocation rule and the magnitude of the savings.

Final interpretation is driven by the behavior of the effect size ratio ρ . When $\rho=1$, the endpoints are equally difficult, so equal splitting is already balanced and all three allocation rules give nearly identical results. As ρ grows farther from 1, however, this means one endpoint begins to determine the study size. In this case, equal splitting becomes less effective because it does not account for the unequal difficulty of the tests. The purpose of the numerical study is therefore to demonstrate how different methods, especially the minimax allocation, redistribute the error budget toward the more difficult endpoint to reduce the overall sample size required for the shared sample design.

7. Results

7.1 Allocation patterns under minimax

The first major pattern is that the minimax rule reallocates error toward the harder endpoint.

Figure 1 below depicts the optimal alpha shares in a two endpoint setting. When $\rho < 1$, test 1 is harder and receives most of α . When $\rho > 1$, test 2 becomes harder and receives most of α instead. At $\rho = 1$, the two curves cross at the equal split line, confirming that equal allocation is optimal only in the symmetric case.

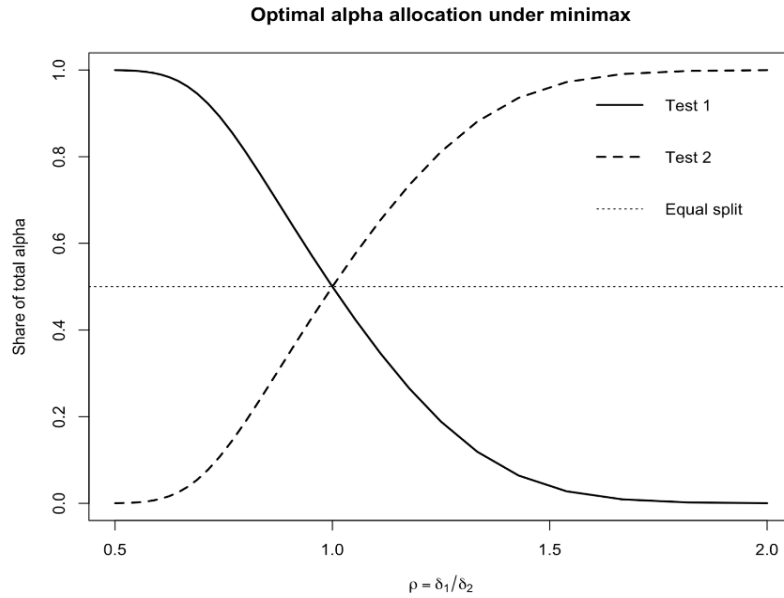


Figure 1. Optimal allocation of alpha under minimax error spending between two endpoints

Additionally, the beta allocation shows the same behavior. The minimax rule is not merely shifting Type I error while leaving power unchanged. It reallocates both α and β so that the hardest endpoint is given enough flexibility to stop dominating the overall design.

7.2 Sample size savings relative to standard Bonferroni

The second major pattern is that sample size savings are smallest near $\rho = 1$ and grow as heterogeneity increases. The proportional saving plot in Figure 2 shows that the reduction

$1 - N_{OPT}/N_{BONF}$ drops to zero at $\rho = 1$ and increases as the effect sizes separate. This matches the theoretical expectation: if the tests are identical, there is nothing to gain from reallocation, but if they differ, the design becomes more improvable.

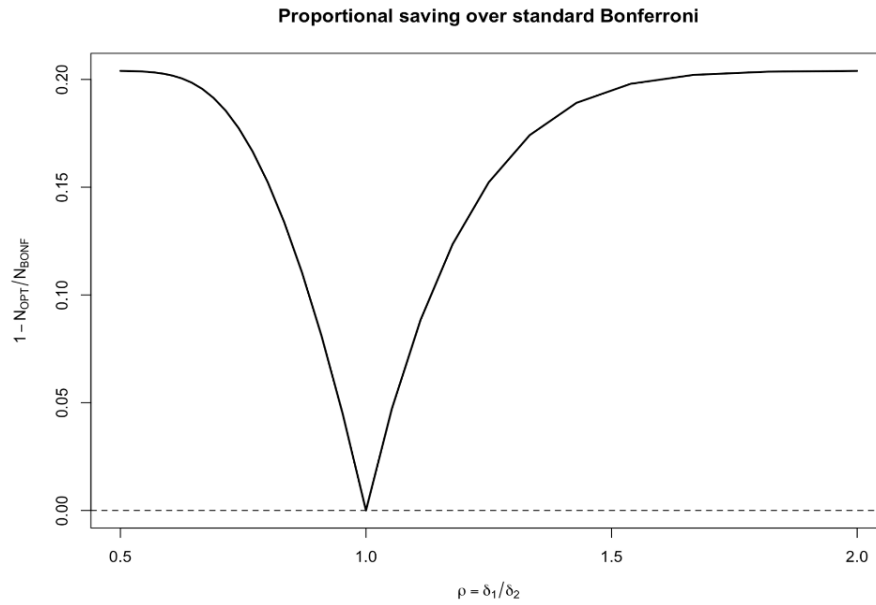


Figure 2. Proportional sample size saving $N_{BONF} - N_{OPT}$ as a function of $\rho = \frac{\delta_1}{\delta_2}$

7.3 Absolute sample size savings relative to standard Bonferroni

A similar pattern appears in the absolute sample size savings shown in Figure 3. The difference $N_{BONF} - N_{OPT}$ is equal to zero at $\rho=1$, where the endpoints are equally difficult and no improvement from reallocation is possible. As ρ moves away from 1, the absolute saving increases, indicating that minimax allocation reduces the total required study size more substantially as endpoint heterogeneity grows. This confirms that the benefit of optimal error spending is largest in the scenarios where one endpoint begins to dominate the experiment.

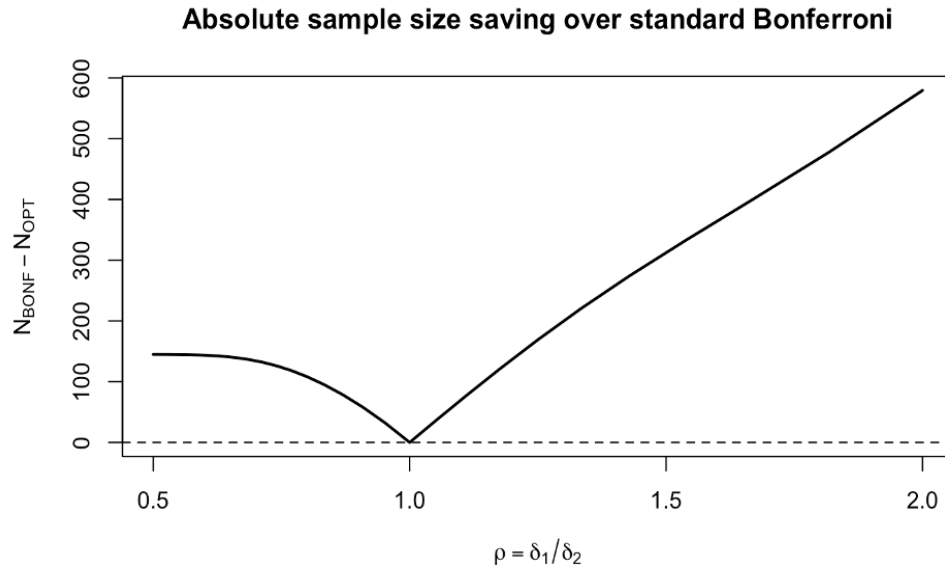


Figure 3. Absolute sample size saving $N_{BONF} - N_{OPT}$ as a function of $\rho = \frac{\delta_1}{\delta_2}$

7.4 Worked example comparison

Table 1 shows a worked two endpoint example recreating the comparison. It shows that both proportionate and minimax allocation put most of the α and β on the harder endpoint, significantly reducing the required overall N relative to equal Bonferroni splitting.

Method	Overall N	Alpha 1	Alpa 2	Beta 1	Beta 2
Bonferroni	262.05	.025	.025	.0750	.0750
Proportionate	164.85	1.2e-05	.049988	.00343	.14656
Minimax	163.46	1.8e-04	.049821	.000429	.14957

Table 1. Comparison of Bonferroni, Proportionate, and Minimax error spending for overall N and error allocations

The numerical findings of Table 1 show that equal Bonferroni splitting requires about 263 observations, whereas the proportionate and minimax rules both reduce the required size to about 164 and 165. Minimax is slightly better than proportionate and achieves the smallest overall N while satisfying the same total Type I and Type II constraints. The allocation columns show exactly how that happens. Almost all of α and β are sent to the more difficult endpoint, while the easier endpoint receives a very small share of the budget.

7.5 Direct comparison of overall required sample size

The final plot, Figure 4, directly compares standard Bonferroni and minimax overall sample sizes across values of ρ . The two curves meet when $\rho = 1$ and separate as heterogeneity grows, with the minimax curve staying below the Bonferroni curve whenever the endpoints are unequal. This provides a visual summary of the main result of the project: optimal reallocation is most beneficial when endpoint difficulties differ and results in a smaller required overall sample size.

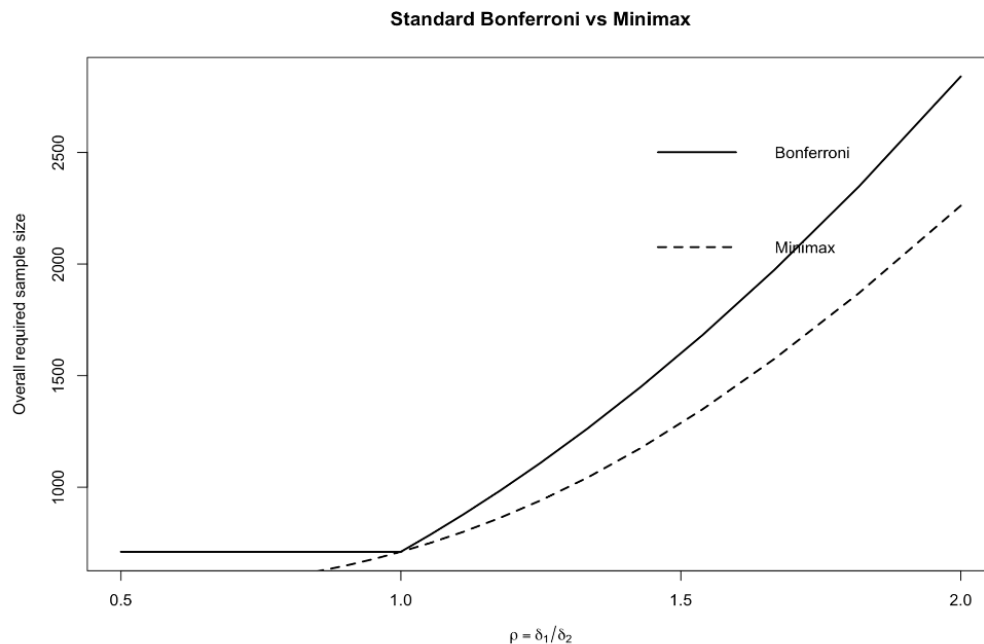


Figure 4. Graph comparing the overall required sample size under Bonferroni versus minimax allocation.

8. Discussion

The main conclusion of the numerical study is that equal Bonferroni splitting is not optimal once endpoints differ in difficulty. In the shared sample setting, the overall study size is determined by the single endpoint with the largest required sample size. Because of that, proper allocation should pay attention to the hardest test rather than enforce symmetry. The minimax rule does exactly this. By shifting more Type I and Type II errors to the harder endpoint, it balances the specific endpoint sample sizes and reduces the maximum within them. This does not mean the design becomes weaker or less accurate. The total familywise budgets are still held but the method simply uses them more efficiently. The result is a smaller study that preserves the same overall error control.

The worked example shows that the savings can be large in practice. A reduction from 263 to about 164 observations is not a minor improvement. It suggests that optimal allocation can meaningfully reduce recruitment burden and study cost when endpoint heterogeneity is substantial. That is why the method is especially ideal for clinical trials, where multiple endpoints are common and participant enrollment is expensive (Dmitrienko, Tamhane, & Bretz, 2010).

Another important point is that larger samples are not automatically better if the design is inefficient. What matters is whether the study has enough observations to meet the desired error and power requirements. If those requirements can be met with fewer participants through better error allocation, then the smaller design is not a compromise in quality, it is a gain in efficiency.

9. Conclusion

This project implemented and interpreted generalized Bonferroni and minimax error spending for multiple endpoint testing. Using Dr. Baron's R framework, it compared equal Bonferroni splitting, proportionate spending, and minimax optimal spending in the common case where one overall sample must satisfy all endpoints.

Evidence from graphs and numerical computations support three final conclusions. First, equal splitting is optimal only when endpoint effect sizes are equal. Second, when endpoints differ, the minimax rule reallocates both α and β toward the more difficult test. Third, the redistribution reduces the overall required sample size, sometimes by a substantial amount, while preserving the same total familywise Type I and Type II error constraints.

Overall, the minimax method provides a more efficient testing strategy whenever endpoint difficulties are unequal. By allocating error probabilities more carefully across endpoints, it can reduce the total sample size required for the study while maintaining the desired error control. In the context of clinical trials, by reducing the number of participants required, this can lower study costs and shorten the time needed to complete the trial and analysis, making it a practical and valuable design tool for modern trials.

References

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